

featuretree: Verified Recovery of Editable CAD Feature Trees from STEP, for Open CAD Systems

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<https://github.com/punkfab/featuretree>

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Abstract

Importing a STEP file into a parametric CAD system yields a single frozen solid: the design history is gone, and no pad depth or hole diameter can be edited. Recovering that history is a well-studied problem — the acceptance criterion we use was stated as a definition by Gupta and Nau [2] in 1995, and commercial implementations have shipped since roughly 1998. It is nonetheless absent from every open-source CAD system and from every callable library we could find: the capability exists only inside proprietary seats costing thousands of dollars per year, with no API. We describe **featuretree**, an MIT-licensed, training-free recogniser that recovers an editable feature tree from an exact B-rep, gates every recovery by re-executing the tree and comparing the result against the input, and emits into FreeCAD and Onshape — two systems with no import-time feature recognition at all. On the eleven-part NIST MBE PMI conformance corpus it returns VERIFIED for 3 parts and PARTIAL for 8, producing *no silently incorrect trees*: all three VERIFIED recoveries exceed 99.68 % IoU against the input, while every PARTIAL one scores at most 74 %. We also report a negative result about our own acceptance criterion: a scalar volume comparison lets over-cut and uncut material cancel, and on one part a reconstruction agreeing to 2.43 % on volume matches no orientation of the target’s bounding box at all. A two-sided IoU test eliminates the failure mode. We claim no methodological novelty; we claim an available, honest implementation and a measured account of where it fails.

1 Introduction

A solid model in a neutral exchange format carries geometry but no design history. Opened in a parametric CAD tool it is one frozen body. Recovering an editable history from such a body is the classical problem of automated feature recognition (AFR), and it is genuinely old: Gupta and Nau [2] defined a recovered feature set as correct precisely when subtracting those features from stock reproduces the part, and Vandenbrande and Requicha [8] built a rule-based system that rejected invalid candidates and iterated to a complete decomposition. Both date from the mid-1990s.

The capability also ships. SolidWorks FeatureWorks has recognised features on imported bodies since approximately 1998, and PTC’s Creo Feature Recognition Tool, available since Pro/ENGINEER Wildfire 4.0, states in its documentation that it “validates the model to ensure that the new features do not modify its geometry” [5]. Geomagic Design X recovers a parametric tree and transfers it into five different CAD systems. The ISO 10303 standard went further still:

Parts 55, 108, 111 and 112 standardise construction history, parameters and constraints, rolled into AP203 edition 2 and AP242 [4].

So why can you not do this? Every implementation above is a GUI feature inside a proprietary seat, priced from thousands to tens of thousands of dollars per year, and none exposes an API a program can call. The two CAD systems most accessible to an independent developer have nothing: Onshape’s documentation states that files imported from another system “do not have a feature tree in Onshape”, and FreeCAD has carried feature recognition as an open request since 2011, with maintainers reporting no capacity to implement it. The standardised route fared no better — Pratt and Kim [4] reported in 2006 that “there are at present no commercial STEP translators making use of” the construction-history capabilities, and we are not aware of published work since that revisits the question. Twenty years later the practical situation is unchanged.

This paper is a response to that gap, not to a gap in the methods literature.

Contributions.

1. **An open implementation (§3).** A training-free, MIT-licensed recogniser from exact B-rep to an editable feature tree, emitted natively into FreeCAD PartDesign and Onshape, and to a `build123d` solid, from one neutral intermediate representation. It is a library, not a GUI feature.
2. **An evaluation with an explicit verdict per part (§4).** On the NIST MBE PMI conformance corpus we report VERIFIED/PARTIAL/REFUSED per part, with reconstruction IoU for *every* part — bounded above for the ones that failed, which is the direction that matters. We are not aware of another AFR evaluation that reports a refusal class alongside a success rate, or that quantifies how wrong its failures are.
3. **A negative result about the acceptance criterion (§5).** Over-cut and uncut material enter a volume difference with opposite signs. We measure the resulting cancellation and show why a two-sided test is required.

What this paper does not claim. It does not claim novelty for verification-gated recovery, for neutral feature representations, for multi-target emission, or for round-tripping GUI edits. Each is anticipated; §2 says by whom. The claim is availability, measurement, and honesty about failure.

2 Related work, and what it anticipates

We state the anticipating work plainly, because the contribution here does not depend on it being absent.

The acceptance criterion is not new. Gupta and Nau [2] require that subtracting the recovered features from stock reproduce the part exactly, and return a binary verdict — a part with no feasible plan “is considered unmachinable”. Ours is the same gate with a measured tolerance in place of set identity. In the learning literature, Willis et al. [9] accept a reconstruction when $\text{IoU} = 1$, noting explicitly that a recovered sequence need not match the ground-truth sequence: geometric equivalence rather than intent recovery, published in 2021.

Neutral feature representations are not new. The macro-parametric approach [3] defines tool-agnostic modelling commands replayed into a target system, and Proficiency’s commercial Universal Product Representation [6] held both parametric and B-rep versions of each feature and regenerated native trees in six CAD systems from around 2000.

Multi-target emission and GUI round-trip are not new. The BYU Neutral Parametric Canonical Form line [7] supports concurrent editing across NX, CATIA and Creo, capturing edits made in each GUI and propagating them through a neutral database. It keys on database GUIDs where we key on stable feature names — an implementation difference, not an inventive one.

Query-based references are not new. Cascaval et al. [1] formalise references as “queries that select elements from the constructed geometry”, exactly the alternative to stored kernel identifiers that we adopt, with correctness criteria we do not attempt.

What we did not find. We searched for, and did not find: any open-source B-rep-to-feature-tree recogniser; any AFR evaluation reporting a graded verdict with a measured geometric residual per part; and any published analysis of why the standardised AP242 construction-history route was not adopted. Absence in one search is weak evidence, and we report these as unfound rather than as nonexistent.

3 System

The intermediate representation is a list of named, ordered features — `sketch`, `pad`, `pocket`, `revolve`, `prism_cut`, `fillet` — as plain JSON. Geometry is selected by *query*, re-resolved against live geometry at each build rather than by persisted kernel identifiers, which is what lets one IR drive structurally different backends and lets edits be matched back by name. Authored IR uses human-meaningful names; recovered IR uses generated ones (`pocket0`, `cut2`), stable enough for the round trip but carrying no design intent.

Recovery proceeds cross-sectionally. Candidate axes are identified along which the solid is prismatic, or about which it is a body of revolution; the outline and through-holes are taken from cross-sections sampled at several heights; blind features are deferred rather than drilled speculatively, since over-cutting cannot be undone; and when the base extrude does not verify, the residual is carved feature by feature, each remaining lump itself recognised as a profile extruded along its own axis. Because several axes can appear prismatic, candidates are not scored heuristically — one is generated per axis and the verifier selects the winner.

The gate. Let S be the input solid, T a candidate tree and $E(T)$ its re-execution. We accept when the volume difference and a rotation-tolerant bounding-box comparison both fall within tolerance, and report PARTIAL with the residual otherwise. When no axis yields a recognisable extrude or body of revolution, the recogniser REFUSEDs outright rather than returning a best guess. Section 5 shows why this particular invariant is weaker than it looks.

4 Evaluation

Corpus. The NIST MBE PMI Validation and Conformance Testing artifacts, “AP203 geometry only” variants: eleven parts (CTC-01–05, FTC-06–11), pure B-rep with no PMI annotations. These are human-designed conformance artifacts intended to be difficult. NIST’s own documentation notes that they are not error-free reference files and that conformance checkers report syntax errors in some of them, so a poor recovery may reflect the input rather than the recogniser.

Table 1: Recovery over the eleven-part NIST corpus. IoU is computed after rotation-aware co-registration, because recovery works in the part’s own frame and the recovered solid is generally rotated relative to the input. [†] marks a *lower bound*: a PARTIAL reconstruction has the wrong bounding box by construction, so the extent prefilter admits no orientation and all 24 are scored under a bbox-corner alignment that is not optimal. The true best-aligned overlap can only be higher. Regenerate with `paper/figures/run_corpus.py` and `paper/figures/iou_corpus.py`.

Part	Verdict	$ \Delta V /V$ (%)	Features	Time (s)	IoU (%)
CTC-01	VERIFIED	0.14	54	24.5	99.860
CTC-02	PARTIAL	62.19	64	90.0	3.79 [†]
CTC-03	PARTIAL	41.46	16	5.1	58.54 [†]
CTC-04	PARTIAL	2.43	270	81.5	25.17 [†]
CTC-05	PARTIAL	51.80	106	11.1	11.42 [†]
FTC-06	PARTIAL	31.85	44	11.5	22.86 [†]
FTC-07	PARTIAL	63.41	52	17.7	15.37 [†]
FTC-08	PARTIAL	32.61	24	62.7	27.93 [†]
FTC-09	VERIFIED	0.19	80	5.3	99.684
FTC-10	PARTIAL	20.80	24	26.3	74.35 [†]
FTC-11	VERIFIED	0.00	2	2.9	100.000

Results. Three parts VERIFIED (27 %) and eight PARTIAL (73 %); no part was refused.

The result we consider most important is that **no incorrect tree was accepted**. Every VERIFIED recovery exceeds 99.68 % IoU against the input. A 27 % acceptance rate paired with an explicit, machine-readable verdict is a different and more useful contract than a higher rate with no soundness signal, and it is the property that makes the output safe to edit.

What the partial IoU figures do and do not say. They are lower bounds, and the direction matters. We search 24 axis-aligned orientations under a bounding-box-corner alignment, so the true best-registered overlap is at least the figure reported and may be higher. Consequently these numbers establish a *floor* on how much of each part was recovered — 3.79 % at worst, 74.35 % at best — and we explicitly do *not* claim from them that the PARTIAL recoveries are geometrically poor, nor that the two verdict classes are separated. Establishing that would need proper registration, which we have not done. What the figures are sound for is the opposite inference: a high lower bound certifies that a recovery is largely right, and FTC-10 at 74.35 % is the closest of the eight.

An earlier threshold, not the vocabulary, was the binding constraint. A previous version of this evaluation refused CTC-02, FTC-07 and FTC-10 outright. Those refusals turned out to be artifacts of two conservative choices in the front end rather than of the feature vocabulary. First, candidate extrude axes were discarded when more than 40 % of faces were non-conforming; threaded and heavily filleted parts carry so much torus/cone/sphere area that they were rejected before their profile was ever examined. Second, a cross-section that was disjoint at every sampled height was treated as a failure, when it in fact describes a part made of several parallel prisms — FTC-07 is two equal lobes — and is recoverable as one pad per region. Relaxing the first to 75 % and adding the second converted all three refusals into PARTIAL recoveries.

Crucially, this was safe to do *because verification is downstream and independent of it*. The axis threshold is a recall knob, not a correctness knob: loosening it can only move a part from REFUSED to PARTIAL or VERIFIED, and the verifier still has to accept the result. We confirmed this empirically — no false accept appeared, and CTC-01’s IoU improved slightly (99.841 % to 99.860 %) — but the property is structural, and it is the practical argument for separating recognition from acceptance. Refusal remains reachable for geometry outside the vocabulary entirely (a torus is still refused); it simply is not exercised by this corpus.

Scale and comparability. This is eleven parts. It establishes that the verdict contract holds on non-trivial standard geometry; it does not establish a rate over a part distribution, and no general recovery rate should be read into it. FTC-11 in particular is a two-feature revolve recovered in 1.6s and should carry little weight. We note explicitly that these numbers are *not* comparable to the 99 %-plus accuracies reported on MFCAD++ and similar benchmarks: those measure per-face classification against ground-truth labels on procedurally generated data, whereas this measures whether a recovered program rebuilds a human-designed conformance part.

5 A negative result: volume comparison admits cancellation

The scalar $|\Delta V|/V$ conceals its own structure. On CTC-01 the recovered solid is *smaller* than the original ($\Delta V = -20\,335\text{ mm}^3$), so the residual cannot be attributed to uncut chamfers, which would make it larger. Co-registering the solids and decomposing by inclusion–exclusion gives $20\,397\text{ mm}^3$ of *over-cut* against 62 mm^3 of *uncut* material, a cancellation factor of 1.01: on this recovery the two-sided discrepancy and the scalar almost coincide.

We report that honestly, because an earlier recovery of the *same part* by an earlier version of this recogniser decomposed as $19\,036\text{ mm}^3$ over-cut against $4\,186\text{ mm}^3$ uncut — a factor of 1.56. The mechanism is real and the magnitude is a property of the particular recovery, not of the part: nothing in the criterion bounds it, and it moved by 50% of its own value under a change to the front end that was not aimed at it. A criterion whose error term varies that freely with unrelated implementation choices is not one to accept results on.

Nothing in a volume-plus-extent test bounds that cancellation factor. A VERIFIED verdict under it should be read as “reproduces the part’s volume and extent to within ϵ ”, not “is the part to within ϵ ”. A two-sided test — IoU, or a symmetric-difference bound — removes the failure mode entirely, because over-cut and uncut material both increase the symmetric difference. IoU is additionally the metric the CAD-program-inference literature already reports, so adopting it buys comparability at the same time. We report IoU in Table 1 for this reason, and recommend it as the acceptance criterion in place of the volume test the implementation currently ships.

CTC-04 shows volume proximity is not geometric proximity. The argument above is structural; CTC-04 gives it a part number. That part reconstructs to within 2.43% on volume — the closest of the eight PARTIAL parts, and on the scalar alone it looks nearly recovered. Yet *no* orientation of its reconstruction matches the input’s bounding-box extents to within 0.05 mm: the recovered solid has close to the right amount of material distributed through a measurably different envelope. A volume agreement of 97.6% therefore carries no implication of 97.6% shape agreement.

This is why the shipped gate tests extent alongside volume, and CTC-04 is correctly rejected today at $\epsilon_V = 0.5\%$. The point is what happens under the obvious move when acceptance rates disappoint — loosening ϵ_V to a few percent to admit “nearly right” recoveries. CTC-04 sits inside that relaxation while being demonstrably the wrong shape, and a scalar volume test has nothing left to reject it with. A two-sided criterion is what would make such a relaxation safe to consider at all. We note that our IoU figures are lower bounds and so cannot themselves quantify how wrong CTC-04 is; the bounding-box mismatch is the part of this we can state without qualification.

A practical obstacle. The decomposition had to be derived from the union by inclusion–exclusion, because direct OpenCASCADE difference and intersection between these operands returned degenerate results — an empty intersection for solids whose bounding boxes coincide exactly. Any stronger verifier built on Booleans will meet the same problem.

6 Limitations

- **Eleven parts.** A demonstration of the contract, not a study of a rate. A corpus-scale evaluation remains the obvious next step.
- **partial IoU is a lower bound only.** A wrong recovery has a wrong bounding box, so the extent prefilter admits no orientation and all 24 are scored under a bbox-corner alignment that is not optimal. The PARTIAL figures in Table 1 can therefore only move upward. They are sound as bounds, and they certify a floor on how much of each part was recovered, but they cannot show that a recovery is *bad* — that would need an upper limit, which this method does not provide. Proper registration (ICP or a pose optimiser) would, and is the single change that would most improve this evaluation.
- **Equivalence, not intent.** Verification certifies that $E(T)$ matches S , not that T is the designer’s history. Recovered feature names are generated and carry no design intent.
- **Edge blends are outside the vocabulary.** Fillets and chamfers are not swept profiles and are left as sub-tolerance residual, so a chamfered part can verify while the recovered tree has sharp edges.
- **Recovery is subtractive only.** Additive bosses cannot be recovered; such parts are reported PARTIAL, not approximated.
- **Freeform geometry is rejected, not approximated.** Splines, ellipses, lofts and sweeps admit no faithful sketch-and-pad tree.
- **Tolerances are uncalibrated.** ϵ was chosen to separate observed successes from observed failures on the parts tested. It is not derived from a manufacturing tolerance model.

- **partial failures are not attributed.** We did not separate shortfalls caused by geometry outside the vocabulary from those caused by defects in the input files, which NIST documents as present. With no part refused, this is now the whole of the failure analysis, and it is the largest piece of work this evaluation still owes.

7 Conclusion

The methods here are not new; their availability is. A capability described in the literature in 1995, shipped commercially since 1998 and standardised in AP242 is still absent from open CAD, and that gap is a practical problem for anyone building on FreeCAD, Onshape or a code-CAD toolchain. We have shown that a training-free implementation reaches a useful contract on standard conformance geometry — accepting a quarter of the corpus with no incorrect acceptances, and labelling the rest rather than guessing — and we have measured a real weakness in the acceptance criterion we began with, together with the two-sided replacement that removes it. That weakness is not hypothetical: the part that comes closest to passing on volume matches no orientation of its target’s bounding box.

Availability

`featuretree` is MIT-licensed and has been publicly available since 2026-06-26 at <https://github.com/punkfab/featuretree>; the recognition component dates from 2026-07-10. Every number reported here is reproduced by the commands in `paper/figures/README.md`, verified against the repository at commit `f7fce78` (2026-09-28). **Note that this is later than the tagged v0.1.0 release:** the evaluation in this version reflects the multi-region and axis-gate changes described in §4, which postdate that tag, so v0.1.0 does not reproduce these figures. The prior-art scan supporting §2 is in `paper/prior-art/FINDINGS.md`.

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